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A MULTIGRID ALGORITHM FOR THE p-LAPLACIAN*

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Abstract. We introduce a full approximation storage (FAS) multigrid algorithm to find the finite element solution for a class of nonlinear monotone elliptic problems. Since the solution of the problem is equivalent to minimize a strictly convex functional, we use a Polak–Ribiere conjugate gradient method as the nonlinear smoother in our algorithm. The advantage in so doing is that we do not have to calculate derivatives of operators. We prove local convergence of our algorithm and illustrate its performance by solving benchmark problems.

Key words. p-Laplacian, nonlinear monotone operators, finite elements, FAS multigrid, Polak–Ribiere conjugate gradient

AMS subject classifications. 65N20, 65N30, 65F10

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1. Introduction.

Let Ω be an open bounded set in $\mathbb{R}^2$ with Lipschitz boundary $\partial\Omega$. Given $p \in (2, \infty)$ and $f \in W^{-1,q}(\Omega)$, with $q = \frac{p}{p-1}$, we consider the Dirichlet problem for the p-Laplacian

$$\begin{cases} -\nabla \cdot (|\nabla u|^{p-2}\nabla u) = f \quad \text{in} \quad \Omega, \\ u|_{\partial\Omega} = 0, \end{cases} \tag{1}$$

where $|\cdot|^2 = (\cdot,\cdot)_{\mathbb{R}^2}$. For $p = 2$ problem (1) reduces to the linear Laplacian. Equation (1) occurs in many mathematical models of physical processes such as glaciology, nonlinear diffusion and filtration (see Philip [14]), power-law materials (Atkinson and Champion [1]), and quasi-Newtonian flows (Atkinson and Jones [2]), for example.

Various numerical algorithms have been proposed to compute the numerical solution to the weak formulation of (1). For instance, Brezis and Sibony [4] analyze a Jacobi-like iterative algorithm. Glowinski and Marrocco [9] use the augmented Lagrangian method to compute the finite element solution of (1). Although the augmented Lagrangian method is robust and better than the method proposed by Brezis and Sibony, its rate of convergence is somewhat slow. Thus, in order to improve the convergence rate of the augmented Lagrangian, Wei [17] proposes a gradient algorithm. More recently, Barrett and Liu [3] apply the Polak–Ribiere nonlinear conjugate gradient method (PRNCG) to compute the numerical solution of the finite element formulation of (1). They report that in their numerical tests the PRNCG performs better than the gradient method.

We propose and analyze in this paper a full approximation storage (FAS) nonlinear multigrid algorithm to compute the finite element solution of (1). Based on the mathematical properties of the boundary value problem (1) our multigrid algorithm uses the PRNCG as a smoother. A standard reference for the numerical analysis

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of the FAS nonlinear multigrid algorithm is Hackbusch [10]. Since the analysis in [10] refers to a general nonlinear problem, then general conditions on the derivative of the discrete operator and the derivative of the relaxation (smoother) operator are assumed. Under such conditions, Hackbusch proves local convergence of the FAS nonlinear multigrid algorithm for a sufficiently fine grid. Later, Reusken [15], [16] also established the local convergence of an FAS multigrid method when it is applied to compute the finite element solution of elliptic, mildly nonlinear problems. Reusken proposes as a smoother a class of suitable Jacobi algorithms.

The organization of the paper is as follows: Section 2 is devoted to mathematical preliminaries. In section 3 we give the finite element formulation, whereas the multigrid algorithm is introduced in section 4. In section 5 we prove the local convergence of the algorithm proposed in the paper. Remarks on the implementation and numerical results are presented in section 6. We finish with conclusions in section 7.

2. Mathematical preliminaries. In order to state the weak formulation of (1) we consider the operator

$$A : W_0^{1,p}(\Omega) \longrightarrow W^{-1,q}(\Omega)\,,$$

$q = \frac{p}{p-1}$, defined by

$$\langle Au, v\rangle = \int_\Omega |\nabla u|^{p-2}\nabla u \cdot \nabla v \, d\Omega\,, \quad \forall u, v \in W_0^{1,p}(\Omega),$$

where $\langle\cdot,\cdot\rangle$ denotes the duality pairing, $W_0^{1,p}(\Omega)$ is the usual Sobolev space on Ω whose elements vanish on the boundary $\partial\Omega$, and $W^{-1,q}(\Omega)$ is its dual space.

Throughout this paper we use the notation $W^{m,p}(\Omega)$ for the Sobolev spaces with norm

$$\|v\|_{m,p} = \left(\sum_{|\alpha|\le m} \|D^\alpha v\|_{L^p(\Omega)}\right)^{1/p},$$

for $1 \le p < \infty$, and

$$\|v\|_{m,\infty} = \max_{|\alpha|\le m} \|D^\alpha v\|_{L^\infty(\Omega)}.$$

We note that the seminorm

$$|v|_{m,p} = \left(\sum_{|\alpha|= m} \|D^\alpha v\|_{L^p(\Omega)}\right)^{1/p}$$

is equivalent to the norm $\|v\|_{m,p}$ on $W_0^{1,p}(\Omega)$.

It is well known (see [6]) that for $p > 2$ the operator A is uniformly monotone and continuous, i.e., there exist positive constants C_1 and C_2 such that, for $u, v \in W_0^{1,p}(\Omega)$,

$$\langle Au - Av, u - v\rangle \ge C_1 \|u - v\|_{1,p}^p, \tag{2}$$

and

$$\|Au - Av\|_{-1,q} \le C_2 \|u - v\|_{1,p} (\|u\|_{1,p} + \|v\|_{1,p})^{p-1}\,, \tag{3}$$

where $\|\cdot\|_{-1,q}$ denotes the norm of $W^{-1,q}(\Omega)$.

Further, A is also the gradient of the strictly convex functional

$$\begin{aligned} J^* : W_0^{1,p}(\Omega) &\longrightarrow \mathbb{R}, \\ J^*(u) &= \frac{1}{p}\int_\Omega |\nabla u|^p d\Omega. \end{aligned}$$

The weak formulation of (1) is as follows:

$$\text{(P)} \qquad \begin{cases} \text{Find } u \in W_0^{1,p}(\Omega) \text{ such that for all } v \in W_0^{1,p}(\Omega), \\ \langle Au, v\rangle = \langle f, v\rangle. \end{cases}$$

For a proof of the existence and uniqueness of the solution to (P) we referred to [6]. There, it is also established that problem (P) is equivalent to the minimization problem,

$$J(u) = \min_{v \in W_0^{1,p}(\Omega)} J(v),$$

where the functional $J : W_0^{1,p}(\Omega) \to \mathbb{R}$ is defined as

$$J(v) = J^*(v) - \langle f, v\rangle.$$

$J(u)$ is twice Gateaux differentiable and then A is Gateaux differentiable. Next, for all $u, v, w \in W_0^{1,p}(\Omega)$, the following relations hold:

$$\langle Au, v\rangle = J'(u)(v) = \int_\Omega |\nabla u|^{p-2}\nabla u \cdot \nabla v \, d\Omega - \int_\Omega f v \, d\Omega\,; \tag{4}$$

$$\begin{aligned} &\langle DA(u)v, w\rangle = J''(u)(v, w) \\ &= \int_\Omega |\nabla u|^{p-2}\nabla v \cdot \nabla w \, d\Omega + (p-2)\int_\Omega |\nabla u|^{p-4}(\nabla u \cdot \nabla v)(\nabla u \cdot \nabla w)\, d\Omega\,. \end{aligned} \tag{5}$$

Note that $DA(u)$ is a linear operator. Conversely, the application of Holder inequality on (4) and (5) yields the following bounds:

$$|\langle Au, v\rangle| \le C_3 \|u\|_{1,p}^{p-1}\|v\|_{1,p} + \|f\|_{-1,q}\|v\|_{1,p}\,, \tag{6}$$

$$|\langle DA(u)v, w\rangle| \le C_4 \|u\|_{1,p}^{p-2}\|v\|_{1,p}\|w\|_{1,p}\,. \tag{7}$$

3. Finite element formulation. Let Ω_h be a polygonal domain that approximates Ω. Ω_h is divided into a finite number NE of small triangles T_j satisfying the following properties:

(i) $\bar{\Omega}_h = \bigcup_{j=1}^{NE} T_j$.

(ii) For $i \neq j$,

$$T_i \cap T_j = \begin{cases} \emptyset\,, \text{ void, or} \\ P_k\,, \text{ a common vertex, or} \\ \Gamma_{ij}\,, \text{ a common side.} \end{cases}$$

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(iii) Let $h_i = \operatorname{diam}(T_i)$ and $\rho_i = \operatorname{inner}\operatorname{diam}(T_i)$. Then, there exists a constant $\sigma > 0$, such that

$$\frac{h_i}{\rho_i} < \sigma \quad \forall i = 1, \ldots, NE.$$

(iv) Let $\{P_k\}_{k=1}^{M}$, $M > 0$, be the set of vertices in $\bar{\Omega}_h$. If $P_i \in \partial\Omega_h$, then $P_i \in \partial\Omega$.

We define a finite dimensional space associated with the partition $D_h = \{T_i\}_{i=1}^{NE}$ as

$$V = \{v_h \in C(\bar{\Omega}_h) : v_h|_{T_j} \in P_1(T_j), \forall T_j \in D_h\}$$

and

$$V^0 = W_0^{1,p}(\Omega) \cap V,$$

where $P_1(T)$ is the space of linear polynomials defined on T. V has the following approximation properties (see [6]). For $m = 0$ or 1 and for all $T_i \in D_h$, we have

(a) for $q \in [1, \infty]$, $s \in [1, \infty]$,

$$\inf_{\chi \in V} \|v - \chi\|_{m,q,T_i} \le C h^{2(\frac{1}{q} - \frac{1}{s})} h^{2-m} \|v\|_{2,s,T_j} \quad \forall v \in W^{2,s}(T_j),$$

provided $W^{2,s}(T_j) \hookrightarrow W^{m,q}(T_j)$ and $W_2^{2,1}(T_j) \hookrightarrow C(\bar{T}_j)$;

(b) for $q > 2$

$$\inf_{\chi \in V} \|v - \chi\|_{m,q,T_i} \le C h^{1-m} \|v\|_{1,q,T_j} \quad \forall v \in W^{1,q}(T_j).$$

The finite element approximation of (P) is formulated as

$$(\mathrm{P_h}) \qquad \begin{cases} \text{Find } u_h \in V^0 \text{ such that for all } v_h \in V^0, \\ \langle A u_h, v_h \rangle = \langle f, v_h \rangle, \end{cases}$$

and the corresponding minimization problem is then

$$\begin{cases} \text{Find } u_h \in V^0 \text{ such that} \\ J_h(u_h) = \min\limits_{v_h \in V^0} J_h(v_h), \end{cases}$$

where

$$J_h(v_h) = \frac{1}{p} \int_{\Omega_h} |\nabla v_h|^p \, d\Omega - \int_{\Omega_h} f v_h \, d\Omega \,.$$

By virtue of the inclusion $V^0 \subset W_0^{1,p}(\Omega)$, the properties of the functional J are inherited by the corresponding discrete functional J_h.

We note that for $p = 2$, the regularity of u and the finite element error analysis are well established. However, for $p \neq 2$, the regularity of u is less well established because in such a case (1) is a degenerate quasi-linear elliptic problem. In this respect, see [12] and the references cited there-in.

Barrett and Liu [3] have obtained optimal error estimates of u_h for sufficiently regular solutions u achievable for a subclass of data f and Ω. In fact, they prove the following result (see Theorem 4.1 of [3]).

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LEMMA 3.1. *Let $u \in W_0^{1,\infty}(\Omega) \cap W_0^{2,s}(\Omega)$, $s \in [1,\infty]$, be the solution of* (P). *If $|f|^{-\gamma} \in L^1(\Omega)$ for some $\gamma \in (0,\infty)$, or if $|f|^{-1} \in L^\infty(\Omega)$, we set $\gamma = \infty$; then we have, for $r \in [1,p)$, the following error estimate for u_h solution of* ($\mathrm{P_h}$):

$$\|u - u_h\|_{1,r,\Omega_h} \leq \begin{cases} Ch^{2/t} & \text{if} \quad s \geq 2, \\ Ch^{s/t} & \text{if} \quad s \in [1,2), \end{cases}$$

where C is a positive constant independent of h and

$$t = \max\left\{2, r\frac{(s+\gamma)p + (p-2)\gamma s}{(s+\gamma)r + (p-2)\gamma s}\right\}.$$

In particular, for $s \in [1,2]$,

$$\|u - u_h\|_{1,p,\Omega_h} \leq Ch^{s/p}\,.$$

4. The multigrid algorithm. We describe in this section the multigrid algorithm to solve ($\mathrm{P_h}$). In order to simplify the notation, we shall hereafter drop the subscript h from the discrete variables unless otherwise stated.

Let $\{D_k\}_{k=0}^m$ be a sequence of partitions of Ω_h. Given D_0, for $1 \leq k \leq m$, D_k is obtained from D_{k-1} by a regular subdivision that joins the edge midpoints of any triangle by edges to form new triangles so that the discretization parameter h_k of D_k is $h_k = \frac{1}{2}h_{k-1}$. Let V_k^0 be the finite element space associated with D_k; the following inclusion relations hold: $D_{k-1} \subset D_k$ and $V_{k-1}^0 \subset V_k^0$ $\forall k$.

To define the multigrid algorithm we shall introduce auxiliary operators.

First, for every $k = 0, 1, \ldots, m$, we define the mesh-dependent inner product $(.\,,.)_k$ on V_k^0 by

$$(v,w)_k = \sum_{T \in D_k} \int_T vw\, dT\,.$$

The operator $A_k : V_k^0 \longrightarrow V_k^0$ is defined by

$$(A_k w, \Phi)_k = \langle Aw, \Phi\rangle \quad \forall \Phi \in V_k^0$$

and, for every $g \in V_k^0$, the functional $J_k(\cdot\,; g) : V_k^0 \longrightarrow \mathbb{R}$ is defined by

$$J_k(w; g) = \frac{1}{p}\int_{\Omega_h} |\nabla w|^p d\Omega - (g,w)_k \quad \forall w \in V_k^0\,.$$

We also need the fine-to-coarse grid transfer operator $I_k^{k-1} : V_k^0 \longrightarrow V_{k-1}^0$ defined by

$$(I_k^{k-1} w, v)_{k-1} = (w,v)_k \quad \forall w \in V_k^0, \quad v \in V_{k-1}^0,$$

as well as the coarse-to-fine grid transfer operator $I_{k-1}^k : V_{k-1}^0 \longrightarrow V_k^0$, which is defined as the natural injection operator, i.e.,

$$I_{k-1}^k v = v \quad \forall v \in V_{k-1}^0.$$

Note that I_{k-1}^k is the transpose of I_k^{k-1} with respect to inner products $(.\,,.)_k$ and $(.\,,.)_{k-1}$.

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The kth level V-cycle iteration $\mathrm{MG}(k, z_0, g)$ is the approximate solution to

$$A_k z = g,$$

defined as follows.

For $k = 0$, $\mathrm{MG}(0, z_0, g)$ is the solution obtained by application of the Polak–Ribiere algorithm with initial guess z_0 to the minimization problem

$$J_0(u; g) = \min_{v \in V_0^0} J_0(v; g).$$

For $k \geq 1$, $\mathrm{MG}(k, z_0, g)$ is obtained recursively in the following steps. (We denote by $\mathrm{PRNCG}(J, z, s)$, s being a positive integer, the output after s iterations of Polak–Ribiere conjugate gradient algorithm applied to functional J with initial guess z; see below.)

(1) *Presmoothing step.* Let

$$z_1 = \mathrm{PRNCG}(J_k(\cdot; g), z_0, s_1). \tag{8}$$

(2) *Correction step.* Let

$$z_2 = z_1 + I_{k-1}^k(\bar{z} - I_k^{k-1}(z_1)), \tag{9}$$

where

$$\bar{z} = \mathrm{MG}(k-1, q, h)$$

and

$$q = I_k^{k-1} z_1,$$

$$h = I_k^{k-1}(g - A_k z_1) + A_{k-1} q.$$

(3) *Postsmoothing step.* Let

$$z_3 = \mathrm{PRNCG}(J_k(\cdot; g), z_2, s_2). \tag{10}$$

Here s_1 and s_2 are positive integers; $s_1 > 0$ while s_2 may be zero.

The kth level iteration is applied in the strategy of full multigrid algorithm. Denoting by f_k the L^2-projection of f in V_k^0 we have the following.

The Full Multigrid Algorithm.

For $k = 0$,

$$u_0 = \mathrm{MG}(0, v_0, f_0). \tag{11}$$

For $k \geq 1$, the approximate solutions u_k are obtained recursively from

$$\begin{aligned} z &= I_{k-1}^k(u_{k-1}), \\ u_k &= \mathrm{MG}(k, z, f_k). \end{aligned}$$

Some comments are in order. We use the PRNCG as a smoother for various reasons. The first one is due to the equivalence between the minimization of the functional $J_k(v; f_k)$ and the weak solution of problem $(\mathrm{P_h})$. The second reason has to do with the difficulties in accurately computing the discrete derivative of the finite element formulation if one wishes to apply nonlinear relaxation methods such as the Newton–Jacobi or the Newton–Gauss–Seidel.

For the convenience of the reader, we describe briefly the Polak–Ribiere algorithm as is implemented in our calculations.

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PRNCG(J, z, s)
Step 0. Put $z^0 = z$. If $J'(z^0) = 0$, stop.
Step 1. Set $j = 0$, $g^0 = h^0 = -J'(z^0)$.
Step 2. Compute λ_j (Brent's method) such that

$$J(z^j + \lambda_j h^j) = \min\{J(z^j + \lambda h^j) : \lambda \geq 0\}.$$

Step 3. Set

$$z^{j+1} = z^j + \lambda_j h^j.$$

Step 4. Compute $J'(z^{j+1})$.
If $J'(z^{j+1}) = 0$, stop;
else, set

$$g^{i+1} = -J'(z^{j+1}),$$
$$h^{j+1} = g^{j+1} + \gamma_j h^j$$

with

$$\gamma_j = \frac{(g^{j+1} - g^j, g^{j+1})}{|g^j|^2}.$$

Step 5. Set $j = j + 1$. If $j < s$ go to Step 2; else, stop.

5. Convergence analysis. The goal of this section is to prove the local convergence of the FAS multigrid algorithm with PRNCG as a smoother, proposed in this paper to calculate the numerical solution of the p-Laplacian with $p > 2$. In order to do so, we shall introduce further notation and some properties of the operator A_k which are needed in our analysis. For this purpose it is convenient to consider the operator A_k as

$$A_k : V_k^0 \longrightarrow V_k^* ,$$

where V_k^* is the dual space of V_k^0 in the sense that for all $u \in V_k^0$

$$\langle A_k u, v\rangle = \langle Au, v\rangle \quad \forall v \in V_k^0 . \tag{12}$$

Hence, we can write the discrete problem as

$$A_k u_k = f_k \in V_k^* ,$$

the exact solution of which is denoted by u_k^*. Note that u_k^* is the finite element approximation to the solution of the continuous problem

$$Au = f .$$

The first property of A_k we wish to point out is given in the following proposition.

PROPOSITION 5.1. *The operator A_k is Gateaux differentiable in V_k^0.*

Proof. For $u, v, w \in V_k^0$ and $t \in \mathbb{R}$ we have

$$\lim_{t\to 0} \left\langle \frac{A_k(u + tv) - A_k u}{t}, w \right\rangle = \lim_{t\to 0} \left\langle \frac{A(u + tv) - Au}{t}, w \right\rangle = \langle DA(u)v, w\rangle .$$

Then A_k is Gateaux differentiable and DA_k is defined by

$$\langle DA_k(u)v, w\rangle = \langle DA(u)v, w\rangle . \qquad \square$$

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Note that for every $z \in V_k^0$, $DA_k(z)$ is a linear operator that satisfies the following inequalities.

LEMMA 5.1. *There exists positive constants c and C such that, for any $z, v, w \in V_k^0$ the following bounds hold:*

(i)

$$\langle DA_k(z)v, v\rangle \geq c \|v\|_{1,p}^p ;$$

(ii)

$$\|DA_k(z)v\|_{-1,q} \geq c \|v\|_{1,p}^{p-1} ; \tag{13}$$

(iii)

$$\|DA_k(z)v - DA_k(z)w\|_{-1,q} \leq C \|z\|_{1,p}^{p-2} \|v - w\|_{1,p} .$$

Proof. (i) Let $y \in V_k^0$ be such that $y = z + tv$, $v \in V_k^0$, and $t \in \mathbb{R}$. From the definition (12) of A_k and (2) we obtain that there exists a constant $c > 0$ such that

$$c \|y - z\|_{1,p}^p \leq \langle Ay - Az, y - z\rangle.$$

Hence,

$$c \|v\|_{1,p}^p \leq \left\langle \frac{A(z + tv) - Az}{t}, v \right\rangle .$$

Now, letting $t \to 0$, the result follows.

(ii)

$$\|DA_k(z)v\|_{-1,q} = \sup_{\|w\|_{1,p} \neq 0} \frac{\langle DA_k(z)v, w\rangle}{\|w\|_{1,p}} \geq \frac{\langle DA_k(z)v, v\rangle}{\|v\|_{1,p}} \geq c \|v\|_{1,p}^{p-1} .$$

(iii) Since $DA_k(z)$ is a linear operator then from (7) the result follows inmediately. Note that $DA_k(z)$ is also the Frechet derivative of A_k in z because it is continuous in a neighborhood of z. ∎

An easy consequence of (13) is as follows.

COROLLARY 5.1. *For any $z \in V_k^0$, $DA_k(z)$ has a continuous inverse $DA_k(z)^{-1}$ defined on its range and satisfies*

$$\|DA_k(z)^{-1}\| \leq C,$$

where C is a constant independent of z.

Proof. Equation (13) and the bounded inverse theorem for linear operators in Banach spaces bounded below give the result. ∎

Another property of A_k that we will use is given in the following.

PROPOSITION 5.2. *The operator $A_k : V_k^0 \to V_k^*$ is a homeomorphism.*

Proof. This result follows from the fact that A_k is a uniformly monotone operator ([5], [13]). ∎

We next formulate the assumptions that we make with respect to the smoothing properties of PRNCG.

The line searching to compute λ_j will stop if

$$|J(z^j + \lambda_j h^j)| \leq \delta.$$

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We have that the following hold (see [7]):

A1. For any $u_k^0 \in V_k^0$ the sequence $\{u_k^j\}$ generated by PRNCG converges to $u_k^* \in V_k^0$ if δ is small enough. Also, $\|h^j\|_{1,p} \to 0$ as $j \to \infty$.

A2. If $\|u_k^0 - u_k^*\|_{1,p}$ is sufficiently small there exists $\nu < 1$ such that the sequence $\{u_k^j\}$ generated by PRNCG converges to $u_k^* \in V_k^0$ as

$$\|u_k^j - u_k^*\|_{1,p} \leq \nu \, \|u_k^{j-1} - u_k^*\|_{1,p} \, .$$

Following the methodology developed in [10] to analyze the convergence of nonlinear multigrid algorithms, we proceed now with the convergence analysis of the two-grid version of the FAS algorithm.

5.1. Convergence of the FAS two-grid algorithm. We first show that FAS two-grid algorithm is well defined. In our case, it suffices that the following consistency condition holds.

PROPOSITION 5.3. *For all $u_k \in V_k^0$ there exists a constant C such that*

$$\|A_{k-1} I_k^{k-1} u_k - I_k^{k-1} A_k u_k\|_{-1,q} \leq C h_{k-1}^{\alpha} \, , \tag{14}$$

where α is the rate of convergence of the finite element solution.

Proof. In order to have A_{k-1} in both terms of $A_{k-1} I_k^{k-1} u_k - I_k^{k-1} A_k u_k$, we introduce the operator

$$P_k^{k-1} : V_k^0 \longrightarrow V_{k-1}^0$$

defined as

$$(A_{k-1} P_k^{k-1} v, \phi)_{k-1} = (A_k v, \phi)_k = \langle Av, \phi \rangle \, ,$$

for all $v, \phi \in V_{k-1}^0$.

Note that, given $f \in W^{-1,q}$, $P_k^{k-1} u_k$ is the finite element solution in V_{k-1}^0 of the problem

$$A\bar{u} = f \, ,$$

whose finite element solution in V_k^0 is u_k.

It easy to see that

$$\langle A_{k-1} P_k^{k-1} u_k, \phi \rangle = \langle I_k^{k-1} A_k u_k, \phi \rangle \quad \forall \phi \in V_{k-1}^0 \, .$$

So, it follows that

$$\|A_{k-1} I_k^{k-1} u_k - I_k^{k-1} A_k u_k\|_{-1,q}$$

$$= \|A_{k-1} I_k^{k-1} u_k - I_k^{k-1} A_k u_k\|_{V_{k-1}^*} = \|A_{k-1} I_k^{k-1} u_k - A_{k-1} P_k^{k-1} u_k\|_{V_{k-1}^*}$$

$$= \sup_{\phi \in V_{k-1}^0 \, , \, \phi \neq 0} \frac{\langle A_{k-1} I_k^{k-1} u_k - A_{k-1} P_k^{k-1} u_k, \phi \rangle}{\|\phi\|_{v_{k-1}^0}} \leq C \|I_k^{k-1} u_k - P_k^{k-1} u_k\|_{1,p} \tag{15}$$

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by virtue of (3).

To estimate the last term on the right-hand side of (15) we write

$$\|I_k^{k-1}u_k - P_k^{k-1}u_k\|_{1,p} \le \|I_k^{k-1}u_k - u_k\|_{1,p} + \|u_k - P_k^{k-1}u_k\|_{1,p}\,. \tag{16}$$

A further splitting on the first term of the right-hand side of (16) yields

$$\begin{aligned}&\|I_k^{k-1}u_k - u_k\|_{1,p}\\ &\le \|I_k^{k-1}(u_k - \Pi^k\bar{u})\|_{1,p} + \|\Pi^{k-1}\bar{u} - \bar{u}\|_{1,p} + \|\bar{u} - u_k\|_{1,p}\,,\end{aligned} \tag{17}$$

where $\Pi^k : W_0^{1,p} \to V_k^0$ is the standard L^2-projection defined as

$$(\Pi^k\bar{u}, \phi) = (\bar{u}, \phi) \quad \forall \phi \in V_k^0\,.$$

Note that in (17) we have used

$$I_k^{k-1}\Pi^k\bar{u} = \Pi^{k-1}\bar{u}\,.$$

Assuming that $\bar{u} \in W_0^{1,\infty}(\Omega) \cap W_0^{2,s}(\Omega)$ it follows, by approximation theory and Lemma 3.1, that all the terms on the right-hand side of (17) are bounded by $Ch_{k-1}^{s/p}$ so that

$$\|I_k^{k-1}u_k - u_k\|_{1,p} \le Ch_{k-1}^{s/p}\,.$$

Similarly,

$$\|u_k - P_k^{k-1}u_k\|_{1,p} \le \|u_k - \bar{u}\|_{1,p} + \|\bar{u} - P_k^{k-1}u_k\|_{1,p} \le Ch_{k-1}^{s/p}\,. \qquad \square$$

We are now in a position to prove the convergence of the FAS two-grid algorithm

THEOREM 1. *Let u_k^0 sufficiently close to u_k^* for* A2 *to be valid. Then, for s_1 and s_2 large enough, there exists a contraction number $\mu_k < 1$ such that*

$$\|u_k^* - z_3\|_{1,p} \le \mu_k\|u_k^* - u_k^0\|_{1,p}.$$

Proof. First, following notation in (8), (9), and (10), we proceed to find an expression for the error $u_k^* - z_2$ after the correction step and then to estimate $u_k^* - z_3$. We have that

$$u_k^* - z_2 = u_k^* - z_1 - I_{k-1}^k(z^* - I_k^{k-1}z_1)\,,$$

where z^* verifies

$$A_{k-1}z^* = I_k^{k-1}(f_k - A_kz_1) + A_{k-1}I_k^{k-1}z_1\,.$$

Since $f_k = A_ku_k^*$, it follows that

$$z^* - I_k^{k-1}z_1 = DA_{k-1}(\psi_{k-1})^{-1}I_k^{k-1}DA_k(\psi_k)(u_k^* - z_1)\,,$$

where ψ_{k-1} and ψ_k are intermediate points between z^* and $I_k^{k-1}(z_1)$ and u_k^* and z_1, respectively. Then,

$$\|u_k^* - z_2\|_{1,p} = \|I - I_{k-1}^k DA_{k-1}(\psi_{k-1})^{-1}I_k^{k-1}DA_k(\psi_k)\|\|u_k^* - z_1\|_{1,p}\,.$$

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Using Lemma 5.1 and Corollary 5.1 and by virtue of A2, we take s_1 sufficiently large such that

$$\|u_k^* - z_2\|_{1,p} \leq K\nu^{s_1}\|u_k^* - u_k^0\|_{1,p}\,,$$

with $K\nu^{s_1} < 1$.

Now, let us estimate $u_k^* - z_3$. Taking into account (10), A2 yields

$$\|u_k^* - z_3\|_{1,p} \leq \nu^{s_2}\|u_k^* - z_2\|_{1,p} \leq K\nu^{s_1+s_2}\|u_k^* - u_k^0\|_{1,p}\,.$$

Then, $\mu_k = K\nu^{s_1+s_2} < 1$. ∎

Remark. In the case that u_k^0 is not sufficiently close to u_k^* for the validity of A2, we take s_1 sufficiently large such that we can find after $s_1^{'}$ iterations, $s_1^{'} < s_1$, an $\bar{u}_k^0$ sufficiently close to u_k^* for A2 to be valid. Note that A1 ensures that this is achievable. Then,

$$\|u_k^* - z_3\|_{1,p} \leq K\nu^{s_1-s_1^{'}+s_2}\|u_k^* - \bar{u}_k^0\|_{1,p}\,,$$

with $K\nu^{s_1-s_1^{'}+s_2} < 1$.

5.2. Convergence of the FAS multigrid algorithm. We prove the convergence of the FAS multigrid algorithm via a perturbation argument to the proof of FAS two-grid algorithm. Let $\mu_k < 1$ be the contraction number at level k of FAS two-grid algorithm, and let $\gamma_{k-1} < 1$ be the global contraction number of FAS $(k-1)$-grid algorithm. Going back to section 4 (see (8), (9), and (10)), we have that at the correction step

$$\bar{z} = \mathrm{MG}(k-1, q, h)$$

is an approximation to the exact solution $z^* \in V_{k-1}^0$ of $A_{k-1}z = h$. Thus,

$$z^* = \bar{z} + \delta z\,, \quad \delta z \in V_{k-1}^0\,,$$

and the error after the correction step is now

$$u_k^* - z_2 = u_k^* - z_1 - I_{k-1}^k(z^* - I_k^{k-1}z_1) - I_{k-1}^k\delta z\,.$$

Hence, following the same argument as in the proof of Theorem 1, we have that

$$\|u_k^* - z_2\|_{1,p} \leq \|I - I_{k-1}^k DA_{k-1}(\psi_{k-1})^{-1}I_k^{k-1}DA_k(\psi_k)\|\|u_k^* - z_1\|_{1,p} + \|I_{k-1}^k\|\|z^* - \bar{z}\|_{1,p}\,.$$

However,

$$\|z^* - \bar{z}\|_{1,p} \leq \gamma_{k-1}\|z^* - q\|_{1,p}$$

and

$$A_{k-1}z^* - A_{k-1}q = I_k^{k-1}(A_k u_k^* - A_k z_1)$$

so that

$$\|z^* - q\|_{1,p} \leq C\nu^{s_1}\|u_k^* - u_k^0\|_{1,p}$$

and, hence,

$$\|z^* - \bar{z}\|_{1,p} \leq \gamma_{k-1}C\nu^{s_1}\|u_k^* - u_k^0\|_{1,p}\,.$$

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Collecting the estimates, we have that

$$\|u_k^* - z_2\|_{1,p} \leq (K + C\gamma_{k-1})\nu^{s_1}\|u_k^* - u_k^0\|_{1,p}\,.$$

Now, we have again as in Theorem 1,

$$\|u_k^* - z_3\|_{1,p} \leq \nu^{s_2}\|u_k^* - z_2\|_{1,p} \leq (1 + C\gamma_{k-1})\mu_k\|u_k^* - u_k^0\|_{1,p}\,.$$

We define the global contraction number γ_k as

$$\gamma_k = (1 + C\gamma_{k-1})\mu_k\,.$$

Taking s_1 and s_2 large enough such that $\gamma_k < 1$, we have proved the following.

THEOREM 2. *Let u_k^0 sufficiently close to u_k^* in order for* A2 *to be valid. Then, for s_1 and s_2 large enough, the solution z_3 obtained by FAS k-grid algorithm satisfies*

$$\|u_k^* - z_3\|_{1,p} \leq \gamma_k\|u_k^* - u_k^0\|_{1,p}\,,$$

where $\gamma_k < 1$ is the global contraction number of FAS k-grid algorithm.

5.3. Convergence of the full multigrid. It remains to prove the convergence of FAS multigrid algorithm in the framework of full multigrid. To this end, we assume that the weak solution of (P) is in $W_0^{1,\infty}(\Omega) \cap W_0^{2,s}(\Omega)$ with $s \in [1,2]$. Then, from Lemma 3.1,

$$\|u - u_k^*\|_{1,p,\Omega_h} \leq C^* h_k^{s/p}\,. \tag{18}$$

Let $\hat{u}_k$ be the FAS k-multigrid solution and $e_k = u_k^* - \hat{u}_k$, with e_0 as small as we wish. We have that

$$\|e_k\|_{1,p} \leq \gamma_k\|u_k^* - \hat{u}_{k-1}\|_{1,p}$$

$$\leq \gamma_k\left(\|u_k^* - u\|_{1,p} + \|u - u_{k-1}^*\|_{1,p} + \|u_{k-1}^* - \hat{u}_{k-1}\|_{1,p}\right)$$

$$\leq \gamma_k C\left(h_k^{s/p} + \|e_{k-1}\|_{1,p}\right).$$

By iterating the above inequality we obtain

$$\|u_k^* - \hat{u}_k\|_{1,p,\Omega_h} \leq \bar{C}(\gamma_k, \ldots, \gamma_0)h_k^{s/p}\,, \tag{19}$$

where

$$\bar{C}(\gamma_m, \ldots, \gamma_0) = \sum_{j=0}^{k}\prod_{l=0}^{j}\gamma_l C_l\,,$$

with $C_l = 2^l C^*$.

Hence, setting $k = m$ in (18) and (19) we can formulate the following.

THEOREM 3. *Let u be the solution of problem* (P) *and suppose that $u \in W_0^{1,\infty}(\Omega) \cap W_0^{2,s}(\Omega)$ with $s \in [1,2]$. Assume that FAS k-grid algorithm has a global contraction number $\gamma_k < 1$ for $k = 0, \ldots, m$. Let $\hat{u}_m$ be the solution of problem* ($\mathrm{P_h}$) *obtained via the full multigrid algorithm. Then, there exists a constant C such that*

$$\|u - \hat{u}_m\| \leq Ch_m^{s/p}\,.$$

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6. Implementation and numerical tests. In the application of the algorithm we evaluate the integrals of the functional J_h and those of the weak formulation as follows:

(i)

$$\int_{\Omega_h} |\nabla v_h|^p \, d\Omega \approx \sum_{T \in D_k} \mathrm{WG}(v_h, T)^p \, |T| \,,$$

where $|T|$ is the measure of the triangle T and $\mathrm{WG}(v_h, T)$ is the weighted gradient of v_h computed by the formula (see [11])

$$\mathrm{WG}(v_h, T) = \frac{1}{6} \sum_{i=1}^{N_T} \nabla \left(v_h|_{T_i} \right) + \left(1 - \frac{N_T}{6} \right) \nabla \left(v_h|_T \right),$$

with $T_1, \ldots, T_{N_T}$ being the triangles that share one edge with T.

(ii)

$$\int_{\Omega_h} |\nabla v_h|^{p-2} \nabla v_h \nabla \varphi_j \, d\Omega \approx \sum_{T \in D_k} \mathrm{WG}(v_h, T)^{p-2} \int_T \nabla v_h \nabla \varphi_j \,,$$

where φ_j are the basis functions of V^0.

(iii)

$$\int_{\Omega_h} f \varphi_j \, d\Omega$$

is evaluated by a Gaussian quadrature sufficiently accurate on the finest grid. Then, it is scaled conveniently on coarser grids.

To test the performance of our nonlinear multigrid algorithm, we have applied it to compute the numerical solution of two benchmark problems.

The first one, which has as exact solution

$$v(x, y) = 10x(x-1)y(y-1) \,,$$

is formulated as

$$\left\{ \begin{array}{rcl} -\nabla \cdot (|\nabla u|^{p-2} \nabla u) & = & f \quad \text{in} \quad \Omega = [0,1] \times [0,1], \\ u|_{\partial\Omega} & = & 0, \end{array} \right. \tag{20}$$

where $p = 4$ and $f = -\nabla \cdot (|\nabla v|^{p-2} \nabla v)$.

The numerical solution on the coarsest grid is obtained by PRNCG with a stopping criterion of 10^{-8}. We take as a guess solution, i.e., the function v_0 in (11), the function

$$\overline{u}(x, y) = \frac{p}{p-1} \left(\frac{\|f\|_\infty}{2} \right)^{\frac{1}{p-1}} \left((\sqrt{2})^{\frac{p}{p-1}} - \left(\sqrt{x^2 + y^2} \right)^{\frac{p}{p-1}} \right), \tag{21}$$

which is the solution of the problem

$$\left\{ \begin{array}{rcl} -\nabla \cdot (|\nabla \overline{u}|^{p-2} \nabla \overline{u}) & = & f \quad \text{in} \quad B_{\sqrt{2}} \supset \Omega, \\ \overline{u}|_{\partial B_{\sqrt{2}}} & = & 0, \end{array} \right.$$

TABLE 1
Exact solution in $[0,1]\times[0,1]:\ 10x(x-1)y(y-1)$. *Step in the finest grid* $h=0.01$.

# grids	Nodes coarse grid	Smooth. steps	Rel. CPU Time	Rel. L_2 error
1	9409	-	1	5.3820×10^{-4}
2	2401	3	0.120	5.8290×10^{-4}
3	625	3	0.030	1.0193×10^{-3}
4	169	3	0.022	2.0835×10^{-3}
5	49	3	0.020	8.0863×10^{-3}

where $B_{\sqrt{2}}$ is the ball of radius $\sqrt{2}$ and center at $x=0$. We note that $\overline{u}$ is an upper solution of (20) (see [8]).

Our second test is the nonhomogeneous Dirichlet problem

$$\left\{\begin{array}{rcl} -\nabla\cdot(|\nabla u|^{p-2}\nabla u) &=& 1 \quad \text{in} \quad \Omega=[-1,1]\times[-1,1], \\ u|_{\partial\Omega} &=& v, \end{array}\right. \tag{22}$$

where $p=4$ and

$$v(x,y)=\frac{p}{p-1}\left(\frac{1}{2}\right)^{\frac{1}{p-1}}\left(1-\left(\sqrt{x^2+y^2}\right)^{\frac{p}{p-1}}\right)$$

is the solution of the problem

$$\left\{\begin{array}{rcl} -\nabla\cdot(|\nabla v|^{p-2}\nabla v) &=& 1 \quad \text{in} \quad B_1, \\ u|_{\partial B_1} &=& 0, \end{array}\right.$$

with B_1 being the ball of radius 1 and center at $x=0$.

As in the first test, the numerical solution of (22) is calculated by PRNCG in the coarsest grid with a stopping criterion of 10^{-8}. We take as an initial guess the function $\overline{u}$ as in (21) with $f\equiv 1$.

Table 1 shows the results for the first numerical test. The first row of the table corresponds to the solution computed by PRNCG in the finest grid with its CPU time taken as reference. The second and subsequent rows correspond to solutions computed by the full multigrid algorithm with the number of grids indicated in the first column. Since we know the exact solution u, we estimate the error of the numerical solution as

$$\text{Relative } L^2\text{-error}=\frac{\|u-\hat{u}_m\|_{L^2}}{\|u\|_{L^2}}.$$

These values are written in the last column of the table.

We note that with two grids and three pre- and postsmoothing steps the saving in CPU time is about eight times with respect to that spent by the PRNCG method to compute the solution in the finest grid, whereas the relative error remains almost the same. With three grids the CPU time decreases by a factor of four with respect to the case of two grids. As we go from three to four grids no appreciable savings in CPU time are obtained, but the error increases; although much less than when we move to five grids. In this case the CPU time is almost the same as with four grids.

Table 2 corresponds to the results of the second test with two pre- and post-smoothing steps. The behavior exhibited by the numerical results in this case is similar of that of Table 1 so that we may conclude that in order to maintain an error

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TABLE 2
Exact solution in $[-1,1]\times[-1,1]$ *: Radial solution in* B_1*. Step in the finest grid* $h = 0.025$*.*

# grids	Nodes coarse grid	Smooth. steps	Rel. CPU Time	Rel. L_2 error
1	6561	-	1	5.2624×10^{-4}
2	1681	2	0.23	1.5777×10^{-3}
3	441	2	0.053	2.7786×10^{-3}
4	121	2	0.025	4.7439×10^{-3}

of the same order as in finest grid, the coarsest grid should not be too coarse, as long as the number of smoothing steps is kept reasonably low. Conversely, beyond a number of grids (in our computations this number is four), no appreciable CPU time saving is observed.

7. Conclusions. We have formulated and analyzed a FAS multigrid algorithm, in the frame of the full multigrid, to compute the finite element solution of a class of nonlinear monotone elliptic problems. Based on the mathematical properties of both the continuous and discrete problems, we propose as smoother the Polak–Ribiere conjugate gradient with line searching. The computational advantages of using this type of smoother is that there is no need to calculate derivatives of the discrete nonlinear operator.

Since the natural norm of the solution is the $W^{1,p}$-norm we have done the convergence analysis in such a norm, and we have proved that the multigrid solution converges to the solution of the continuous problem with the same order as the *exact* numerical solution.

From the analysis and from the numerical results, we conclude that if the ratio error/CPU time is wanted to be optimal, the coarsest grid must not be very coarse; that is, there is a largest grid size beyond which the solution obtained by our full multigrid for nonlinear problems may deteriorate.

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